> *"At the tone: seven hours, twenty five minutes, coordinated universal time."*

This announcement is followed by a short 1000 Hz tone. The time announcement is concatenated, or assembled from a variety of digitally stored phrases or enunciated words. All the phrases and words are recordings of a human voice.

A station identification message using the same male voice is broadcast at the beginning and mid-point of each hour. The station ID reads as follows:

> *"National Institute of Standards and Technology time: this is Radio Station WWV, Fort Collins Colorado, broadcasting on internationally allocated standard carrier frequencies of two-point-five, five, ten, fifteen, and twenty megahertz, providing time of day, standard time interval, and other related information. Inquiries regarding these transmissions may be directed to the National Institute of Standards and Technology, Radio Station WWV, 2000 East County Road 58, Fort Collins, Colorado, 80524."*

The WWV station ID announcement lasts about 35 seconds, and is broadcast during the minute following the tone that marks the beginning or midpoint of each hour. For example, the announcement runs at 1100 UTC and 1130 UTC. It is stored in its entirety in electronic memory in the time code generator.

c) Standard Time Intervals

Standard time intervals broadcast on WWV are indicated by various tones of various lengths. Intervals of seconds, ten seconds, minutes, hours, and days are all part of the broadcast. The time intervals are indicated as follows:

- Seconds: bursts of 1000 Hz tone; most are 5 ms long, which simulates the ticking of a clock. These pulses or "ticks" are omitted on the 29th and 59th seconds. The second that begins each minute is indicated by a longer tone.

- Ten seconds: indicated on BCD time code.

- Minutes: 800 ms burst of 1000 Hz tone at the beginning of every minute except the beginning of the hour.

- Hours: 800 ms burst of 1500 Hz tone at the beginning of every hour. Also, the 440 Hz tone is broadcast once each hour, at minute 2 on WWV and minute 1 on WWVH, except for the first hour of each day.

- Days: There is no formal day marker, but the 440 Hz tone described above is omitted during the first hour of each day.

d) Standard Audio Frequencies

In addition to the inaudible standard carrier frequencies, WWV also broadcasts standard audio frequencies during most minutes of every hour. Figure 2.5 is an illustration of the WWV audio signals. Tones of 500 Hz and 600 Hz are broadcast on alternating minutes, and a 440 Hz tone is broadcast once each hour, except the first hour of the day. The tones are inserted in the broadcast for the first 45 seconds of their allocated minutes, and are followed by a silent period and the voice announcement indicating the time (Figure 2.5 inset). The seconds pulses interrupt the tones briefly; each pulse is preceded by 10 ms of silence and followed by 25 ms of silence. The entire "protected zone" around each pulse is shown in Figure 2.6. This arrangement makes the pulse easier to discern; however, the short silent periods around each pulse are nearly unnoticeable.

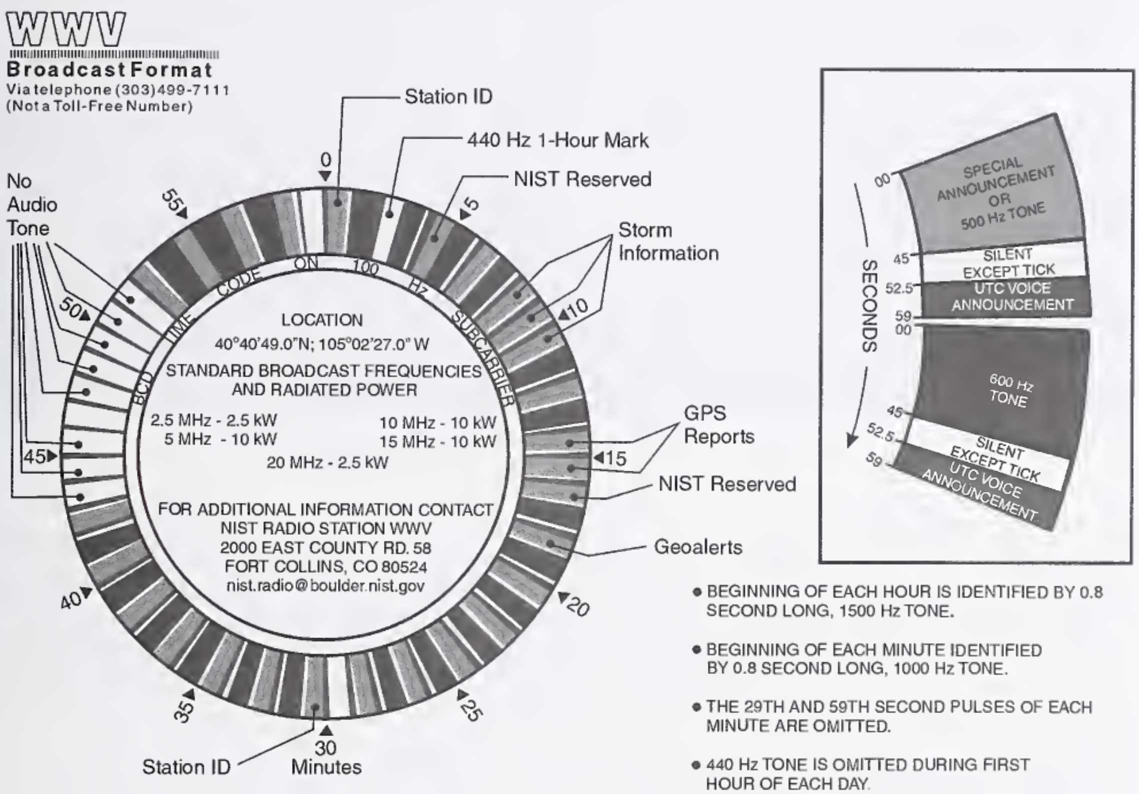

Figure 2.5. WWV audio broadcast format.

Longer silent periods without tone modulation are included in several minutes of every hour of the WWV broadcast format. However, the carrier frequency, seconds pulses, time announcements, and the BCD time code continue during these periods. The purpose of the silent periods on WWV is to prevent interference with voice announcements on the WWVH broadcast. One station will normally not broadcast a tone while the other is broadcasting a voice message. As shown in Figure 2.5, the silent periods for WWV are from 43 to 52 minutes after the hour, as well as minutes 29 and 59.

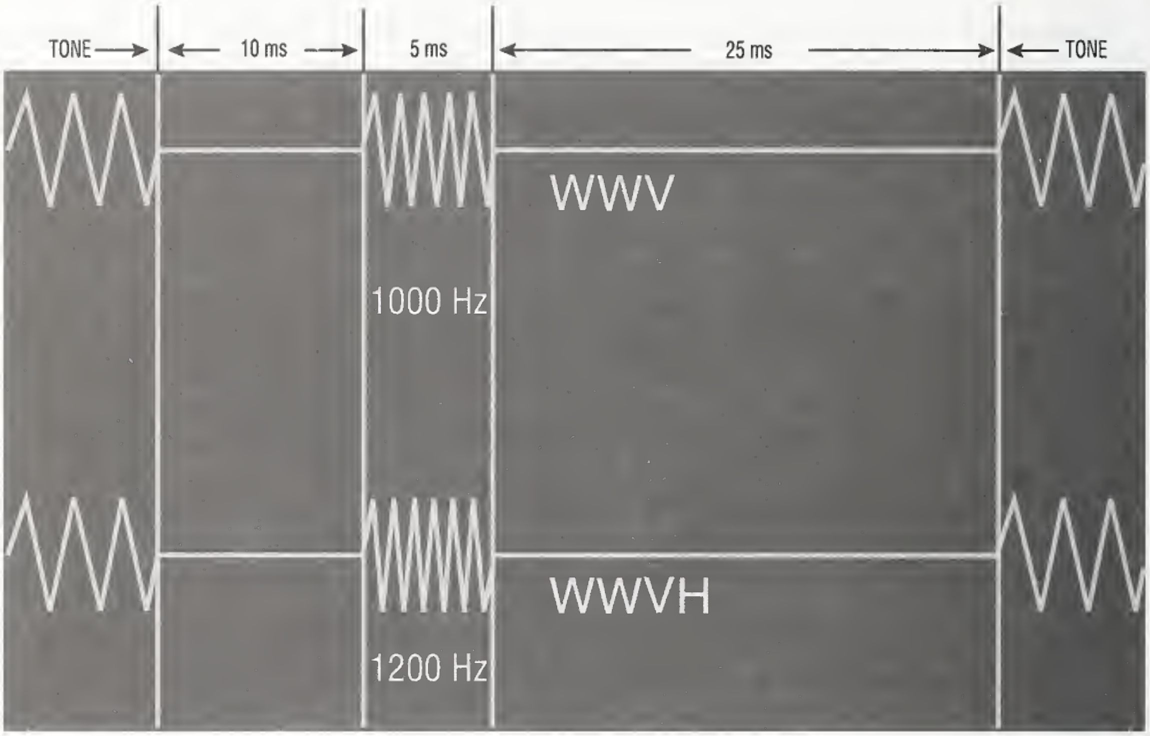

Figure 2.6. WWV and WWVH second pulses.

e) UT1 Corrections

UT1 information is broadcast on WWV during the first 16 seconds of every minute using doubled seconds pulses, or doubled ticks. The amount of correction (in tenths of a second) can be determined by counting the number of doubled ticks. If the ticks occur during the first eight seconds (1 to 8) of the minute, the sign of the correction is positive; if they occur during the second eight seconds (9 to 16), it is negative. UT1 corrections are also broadcast as part of the BCD time code (next section).

f) 100 Hz Time Code

WWV and WWVH broadcast an identical time code on a 100 Hz subcarrier. In addition, the 100 Hz frequency is usable as a standard frequency. It is audible as an intermittent low buzzing sound on most receivers and on the telephone time-of-day service. It originates in the time code generators, and is the only portion of the broadcasts that identify the day of year, and the last two digits of the year.

The time code is in binary coded decimal (BCD) format, where four bits represent one decimal number. The binary-to-decimal weighting scheme is 1-2-4-8, with the least significant bit sent first. The binary groups and decimal equivalents are shown in Table 2.1.

Table 2.1. BCD weighting scheme used by WWV and WWVH time code.

Decimal number	Bit 1 2^0	Bit 2 2^1	Bit 3 2^2	Bit 4 2^3
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	1	1	0	0
4	0	0	1	0
5	1	0	1	0
6	0	1	1	0
7	1	1	1	0
8	0	0	0	1
9	1	0	0	1

The WWV/VH time code is a modified IRIG-H code. IRIG stands for the Inter-Range Instrumentation Group, a consortium of technical personnel representing the different military services, who are responsible for standardizing communications methods at various missile range facilities. In the standard IRIG-H format, the bits are transmitted at a rate of 1 pps on a 100 Hz or 1000 Hz carrier, using amplitude modulation. A binary 0 would be seen as a pulse consisting of increased or "high" modulation of 200 ms duration (20 cycles of a 100 Hz tone) and a binary 1 would see a pulse of 500 ms duration (50 cycles of a 100 Hz tone). Position marker bits are 800 ms duration. The leading positive-going edge of each pulse occurs at the beginning of each second. The ratio of high to low amplitude is 3.3:1 [39].

In the WWV/VH broadcasts, the 100 Hz tone (along with all other tones, see Figure 2.6) is suppressed to allow the seconds ticks to be heard. This has the effect of deleting the first 30 ms of a normal IRIG-H pulse. Thus, on the WWV/VH time code a binary 0 is 170 ms long, a binary 1 is 470 ms long, and a position marker is 770 ms duration. The leading edge of each pulse occurs 30 ms after the beginning of that second.

Each WWV/VH time code frame takes one minute to send. The frame contains the last two digits of the current year, day of year, hour and minute, as well as UT1 corrections, Daylight Saving Time and standard time information, leap second indication, and various identifier and position marker bits. The year (00 through 99), hour (00 through 23) and minute (00 through 59) are each expressed using two BCD groups or sets. The day of year (001 through 366) requires three BCD sets. Seconds can be determined by counting the pulses within the frame. All information in the time code frame is referenced to UTC at the beginning of the frame.

Figure 2.7 shows the WWV and WWVH time code format. Unlike the standard IRIG-H code, no pulse is transmitted during the first second of the minute. This blank space, or "hole," when combined with position marker P_0, constitutes the frame reference marker R, which marks the beginning of a new minute. Since the first 30 ms of each pulse is deleted, the first pulse of the new frame occurs 1.03 s after the beginning of the minute. Position marker bits lasting 770 ms (P_0 to P_9) are inserted every 10 s for synchronization purposes. These position markers also indicate the tens of seconds standard time interval.

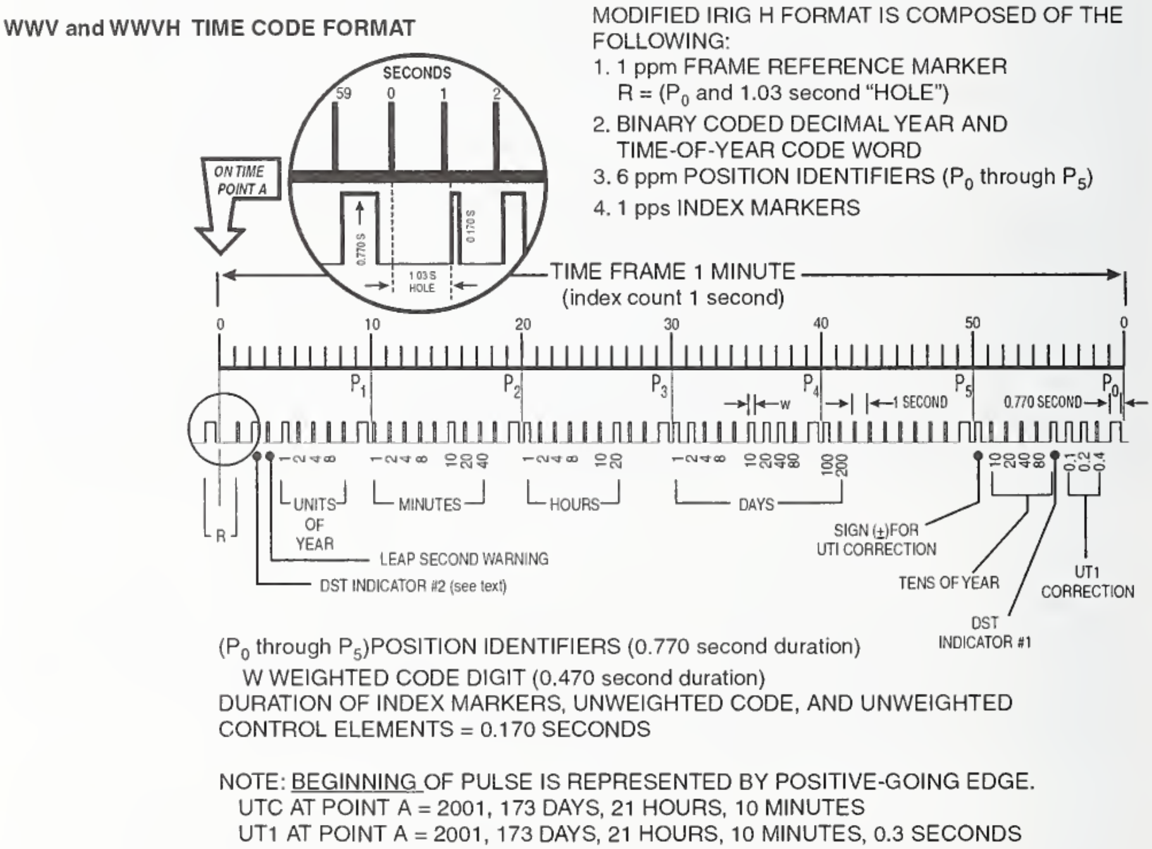

Figure 2.7. WWV and WWVH time code diagram.

WWV and WWVH broadcasts the last two digits of the current year in two separate sections of the time code. Bits 51 through 54 are used to transmit the decade indicator, and bits 4 through 7 are used to transmit the last digit. Therefore, the year 2004 would be indicated by a binary equivalent of decimal 04: binary 0000 on bits 51-54 (decimal 0) and binary 0010 (decimal 4) on bits 4-7. Likewise, the hours are transmitted on bits 20-23 and 25, 26, and the minutes on bits 10-13 and 15-17. Since the day of year function involves three decimal digits, three BCD sets are required: bits 30-33, 35-38, and 40, 41.

UT1 corrections are transmitted on three bits: 56, 57, and 58. This BCD set indicates the amount of correction when multiplied by 0.1. The sign of the correction is found in bit 50; if that bit is high, or set at 1, the correction is added to UTC to find UT1. If bit 50 is low, the correction is subtracted. Since only three bits are used, the UT1 correction range broadcast by WWV/VH is from −0.7 to +0.7 s.

A leap second warning is indicated on the WWV/VH time code at bit 3. If a one is present on bit three, a leap second will be added to UTC at the end of the current month (leap seconds are added only at the end of June or December). The warning bit is set high near the beginning of the month, and immediately after the leap second is added, it is set low, or to zero again.

The WWV/VH time code carries Daylight Saving Time (DST) and standard time (ST) indicators on bits 2 and 55. During standard time, both bits are set low (0). At 0000 UTC on the day ST changes to DST, bit 55 goes high, and bit 2 goes high at 0000 UTC the following day. When DST ends, bit 55 is set low at 0000 UTC the day of the change, and bit 2 goes low 24 hours later.

All unused bits are set to 0 and reserved for future use.

g) Official Announcements

Several minutes of every hour on WWV are allocated for official announcements by U.S. Government agencies. The announcements are of a public service nature and are limited to 45 seconds per segment. More than one segment may be used by a particular agency. Announcement allocations as of this writing can be seen in Figure 2.5 and are described in Table 2.2. For more information about the content and the format of these announcements, see *NIST Special Publication 432* [38], or contact the sponsoring agency.

In addition to the announcements listed in Table 2.2, NIST reserves minutes 4 and 16 for occasional special announcements. These segments are used to relate information of interest to users of the time and frequency broadcasts, such as user surveys or changes to the broadcast. When no announcement is present, a tone is usually broadcast.

The WWV voice storage systems consists of two commercially available voice storage units, manufactured by Racom Corp., and associated filtering and monitoring equipment. Users dial into the voice storage units and leave the messages to be broadcast on an allocated channel in that device. At the appropriate time, the time code generators play that message as part of the composite audio signal sent to the transmitters. All three TCGs receive the audio channel, even though only one is "on the air." A majority logic device prevents erroneous channel triggering.